

Zhichao Peng

CONTACT INFORMATION	Department of Mathematics The Hong Kong University of Science and Technology	pengzhic@ust.hk https://zhichaopengmath.github.io
PROFESSIONAL EXPERIENCE	Assistant Professor, Department of Mathematics, The Hong Kong University of Science and Technology, Aug. 2023 – now Visiting Assistant Professor, Department of Mathematics, Michigan State University, Aug. 2020 – Jul. 2023 Mentors: Prof. Daniel Appelö and Prof. Yingda Cheng Research intern, Los Alamos National Laboratory, May 2019 – Aug. 2019 Mentor: Dr. Xianzhu Tang	
EDUCATION	Ph.D. in Applied Mathematics, Rensselaer Polytechnic Institute, Aug. 2020 Advisor: Prof. Fengyan Li B.S. in Mathematics, Peking University, July 2015	
RESEARCH INTERESTS	• Efficient numerical methods for kinetic equations, acoustic and electromagnetic wave equations in the time and frequency domain • Data-driven and projection based reduced order modeling, especially for kinetic equations and transport problems • Structure preserving numerical methods: asymptotic preserving, positivity preserving, energy stable	
PUBLICATIONS	• Refereed journal papers (published or accepted): 1. W. Guo, <u>Z. Peng</u>, <i>Highly Efficient Rank-Adaptive Sweep-based SI-DSA for the Radiative Transfer Equation via Mild Space Augmentation</i>, accepted by SIAM Journal on Scientific Computing, 2026, arXiv:2603.25233. 2. W. Guo, <u>Z. Peng</u>, <i>An Inexact Low-Rank Source Iteration for Steady-State Radiative Transfer Equation with Diffusion Synthetic Acceleration</i>, Journal of Computational Physics, 569, 115427, 2026, arXiv:2509.00805. 3. L. Ji, <u>Z. Peng</u>, Y. Chen, <i>AAROC: Reduced Over-Collocation Method With Adaptive Time Partitioning and Adaptive Enrichment for Parametric Time-Dependent Equations</i>, Numerical Methods for Partial Differential Equations, 41(5), e70031, 2025, arXiv:2412.02152. 4. <u>Z. Peng</u>, <i>A flexible GMRES solver with reduced order model enhanced synthetic acceleration preconditioner for parametric radiative transfer equation</i>, Journal of Computational Physics, 534, 114004, 2025, arXiv:2410.08735. 5. <u>Z. Peng</u>, <i>Reduced order model enhanced source iteration with synthetic acceleration for parametric radiative transfer equation</i>, Journal of Computational Physics, 517, 113303, 2024, arXiv:2402.10488. 6. <u>Z. Peng</u>, Y. Chen, Y. Cheng, F. Li, <i>A micro-macro decomposed reduced basis method for the time-dependent radiative transfer equation</i>, Multiscale Modeling & Simulation, 22(1), 639–666, 2024, arXiv:2211.04677.	

7. Z. Peng, D. Appelö, S. Liu, *Universal AMG Accelerated Embedded Boundary Method Without Small Cell Stiffness*, Journal of Scientific Computing, 97, article 40, 2023, arXiv:2204.06083.
8. L. A. Martinez, Z. Peng, D. Appelö, D. M. Tennant, N. A. Petersson, J. L. DuBois, Y. J. Rosen, *Noise-specific beating in the higher-level Ramsey curves of a transmon qubit*, Applied Physics Letters, 122(11), 114002, 2023, arXiv:2211.06531.
9. Z. Peng, M. Wang, F. Li, *A learning-based projection method for model order reduction of transport problems*, Journal of Computational and Applied Mathematics, 418, 114560, 2023, arXiv:2105.14633.
10. H. Zhang, Z. Peng, *Total generalized variation for triangulated surface data*, Journal of Scientific Computing, 93, article 87, 2022.
11. Z. Peng, D. Appelö, *EM-WaveHoltz: A flexible frequency-domain method built from time-domain solvers*, IEEE Transactions on Antennas and Propagation, 70(7), 5659–5671, 2022, arXiv:2103.14789.
12. Z. Peng, Y. Chen, Y. Cheng, F. Li, *A reduced basis method for radiative transfer equation*, Journal of Scientific Computing, 91, article 5, 2022, arXiv:2103.07574.
13. Z. Peng, F. Li, *Asymptotic preserving IMEX-DG-S schemes for linear kinetic transport equations based on Schur complement*, SIAM Journal on Scientific Computing, 43(2), A1194–A1220, 2021, arXiv:2006.07497.
14. Z. Peng, Y. Cheng, J.-M. Qiu, F. Li, *Stability-enhanced AP IMEX1-LDG method: energy-based stability and rigorous AP property*, SIAM Journal on Numerical Analysis, 59(2), 925–954, 2021, arXiv:2005.05454.
15. Z. Peng, Q. Tang, X.-Z. Tang, *An Adaptive Discontinuous Petrov–Galerkin Method for the Grad–Shafranov Equation*, SIAM Journal on Scientific Computing, 42(5), B1227–B1249, 2020, arXiv:2001.04524.
16. Z. Peng, Y. Cheng, J.-M. Qiu, F. Li, *Stability-enhanced AP IMEX-LDG schemes for linear kinetic transport equations under a diffusive scaling*, Journal of Computational Physics, 415, 109485, 2020.
17. Z. Peng, V. A. Bokil, Y. Cheng, F. Li, *Asymptotic and positivity preserving methods for Kerr-Debye model with Lorentz dispersion in one dimension*, Journal of Computational Physics, 402, 109101, 2020.

• Preprints:

1. C. Han, L. Einkemmer, W. Guo, Z. Peng, *Low-rank SI-DSA and GMRES-DSA for Radiative Transfer Equation with Adaptive Accuracy Control*, 2026, arXiv:2610.07653.
2. D. Appelö, W. D. Henshaw, Z. Peng, *Numerical Study of Eigenvector Deflation to Accelerate the WaveHoltz Method*, 2026, arXiv:2606.31842.
3. S. Shang, W. Guo, Z. Peng, *Incremental Tensor-Train Compression from Streaming TT-Formatted Data: Applications to Reduced-Order Modeling*, 2026, arXiv:2606.26900.
4. M. Li, Y. Xiang, Z. Peng, *Enhancing Future Prediction of Linear and Non-linear Reduced-Order Models for Transport-Dominated Problems Using Lagrangian Data*, 2026, arXiv:2603.19702.
5. N. Liu, Z. Peng, *On-the-Fly ROM-Based Acceleration of SI-DSA for Implicit Time Marching of the Radiative Transfer Equation*, 2026, arXiv:2603.19647.

6. N. Tang, Z. Peng, *Synthetic Acceleration Preconditioners for Parametric Radiative Transfer Equations based on Trajectory-Aware Reduced Order Models*, 2025, arXiv:2509.05001.
7. A. Galindo-Olarte, Z. Peng, J. K. Ryan, *Superconvergence Extraction of Upwind Discontinuous Galerkin Method Solving the Radiative Transfer Equation*, 2025, arXiv:2509.00296.
8. T. Jin, Z. Peng, Y. Xiang, *Adaptive and hybrid reduced order models to mitigate Kolmogorov barrier in a multiscale kinetic transport equation*, 2025, arXiv:2505.08214.
9. Y.-M. Law, Z. Peng, D. Appelö, T. Hagstrom, *A P-Adaptive Hermite Method for Nonlinear Dispersive Maxwell's Equations*, 2025, arXiv:2504.09269.

• Technical reports:

1. Z. Peng, D. Appelö, N. A. Petersson, M. Motamed, F. Garcia, Y. Cho, *Deterministic and Bayesian Characterization of Quantum Computing Devices*, 2023, arXiv:2306.13747.
2. Z. Peng, D. Appelö, N. A. Petersson, F. Garcia, Y. Cho, *Mathematical approaches for characterization, control, calibration and validation of a quantum computing device*, 2023, arXiv:2301.10712.

AWARDS AND GRANTS

1. GRF 16307226, Towards An Integrated Framework of Low-rank Methods and Data-driven Reduced Order Models for RTE, Sep. 2026–Aug. 2029, PI, 616,160 HKD
2. GRF 16306825, Reduced Order Model Enhanced Preconditioners for Parametric and Time Dependent Radiative Transfer Equation, 2026–2028, PI, 850362 HKD
3. HKUST-KTH Global Knowledge Network Awards 2024, 40000HKD, CO-PI, 2024–2025
4. ECS 26302724, Conquer The Kolmogorov Barrier for Reduced Order Model of The Boltzmann Transport Equation, 2025–2027, PI, 831657 HKD

STUDENTS AND POSTDOCS

- Students: Meng Li (co-supervised with Prof. Yang Xiang, 2023-), Ning Tang (2024-), An Ping (2025-), Chenxi Han (2025-), Shuyuan Shang (2026-)
- Postdoc: Ningxin Liu (2025-)

SERVICES

1. Conference and workshop organization
 - (a) Co-organize “Dimensionality Reduction and Low-rank Methods for PDEs” at Scientific Computing and Differential Equations (SciCADE), Edinburgh, UK, June and July, 2026
 - (b) Co-organize “Algorithms and Theories for the Dimensionality Reduction of Parametric Differential Equations” at The 13th China Mathematical Society Computational Mathematics Annual Conference, Changsha, Hunan, Aug. 2025
 - (c) Co-organize “Recent Developments of Numerical Methods and Modeling for Kinetic and Related Equations” at the Thrid HKSIA Biennial Conference, CUHK, Hong Kong, Jul. 2025
 - (d) Co-organize “Recent Developments in Numerical Methods for Kinetic Equations”, at SIAM CSE 2025, Fortworth, TX, USA, Mar. 2025
 - (e) Co-organize “2024 Workshop on Mathematical Theories and Algorithms for AI for Science” at HKUST, Hong Kong, Sep. 2024

TALKS AT
INTERNATIONAL
AND REGIONAL
CONFERENCES

2. Journal referee:
SIAM Journal on Scientific Computing, SIAM Journal on Numerical Analysis, Journal of Computational Physics, SIAM Journal on Applied Mathematics, Mathematics of Computation, Journal of Scientific Computing, Communications in Computational Physics, Advances in Computational Mathematics, Journal of Applied Mathematics and Physics, IEEE Journal on Multiscale and Multiphysics Computational Techniques
1. The 24th Annual Conference of the China Society for Industrial and Applied Mathematics, Lanzhou, China, Sep. 2026
2. 22nd International Conference for Mesoscopic Methods in Engineering and Science, Hainan University, Haikou, Hainan, China, Aug. 2026
3. China Conference on Scientific Machine Learning, Shanghai Jiaotong University, Shanghai, China, Aug. 2026
4. Seminar talk, Shanghai Tech University, Shanghai, China, Aug. 2026
5. Seminar talk, Shanghai Jiaotong University, Shanghai, China, July 2026
6. 17th World Congress on Computational Mechanics and 10th European Congress on Computational Methods in Applied Sciences and Engineering (WCCM-ECCOMAS), Munich, Germany, July, 2026
7. International Conference on Scientific Computing and Differential Equations, Edinburgh, UK, June and July, 2026
8. The 15th National Conference on Inverse Problems, Imaging and Applications & Annual Meeting of CSIAM Activity Group on Inverse Problems and Imaging, 2026, Wenzhou Kean University, Wenzhou, Zhejiang, China, May 2026.
9. Hong Kong Joint Universities Conference on Structured Matrices and Scientific Computing, Sep. 2025
10. 21st NSNMF, Yining, China, Aug. 2025
11. The Third HKSIAM Biennial Conference, CUHK, Hong Kong, Jul. 2025
12. Advanced Computational Mathematics Workshop, Shanghai Jiaotong University, Shanghai, China, May 2025
13. Efficient Numerical Methods, Beijing Normal University, Zhuhai Campus, Zhuhai, China, May 2025
14. HKUST-KAIST-NUS Joint Workshop on Partial Differential Equations and Scientific Computing, National University of Singapore, Singapore, Apr. 2025
15. SIAM CSE 2025, Fortworth, TX, USA, Mar. 2025
16. Numerical Linear Algebra and Fast Algorithms Youth Forum, Xiangtang, Hunan, China, Jan. 2025
17. Computational Learning for Model Reduction, ICERM, Providence, RI, USA, Jan. 2025
18. First HKMS-HKSIAM Joint Young Scholars Symposium, HKU, Hong Kong, China, Dec. 2024
19. 2024 Workshop on Mathematical Theories and Algorithms for AI for Science, HKUST, Hong Kong, Sep. 2024
20. International Conference on Scientific Computation and Differential Equations, Singapore, July 2024
21. 2024 East Asia Section of SIAM Annual Meeting, Macau, China, June-July 2024

22. 2023 CSIAM Annual Conference, Kunming, China, Oct. 2023
23. 2023 ICIAM, Tokyo, Japan, Aug. 2023